

HDOC Day-43 – C Programming

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Batch-15

Question: Write a menu-driven C program using separate functions to:

1. Find the longest word in a sentence.
2. Check whether one string is a rotation of another.

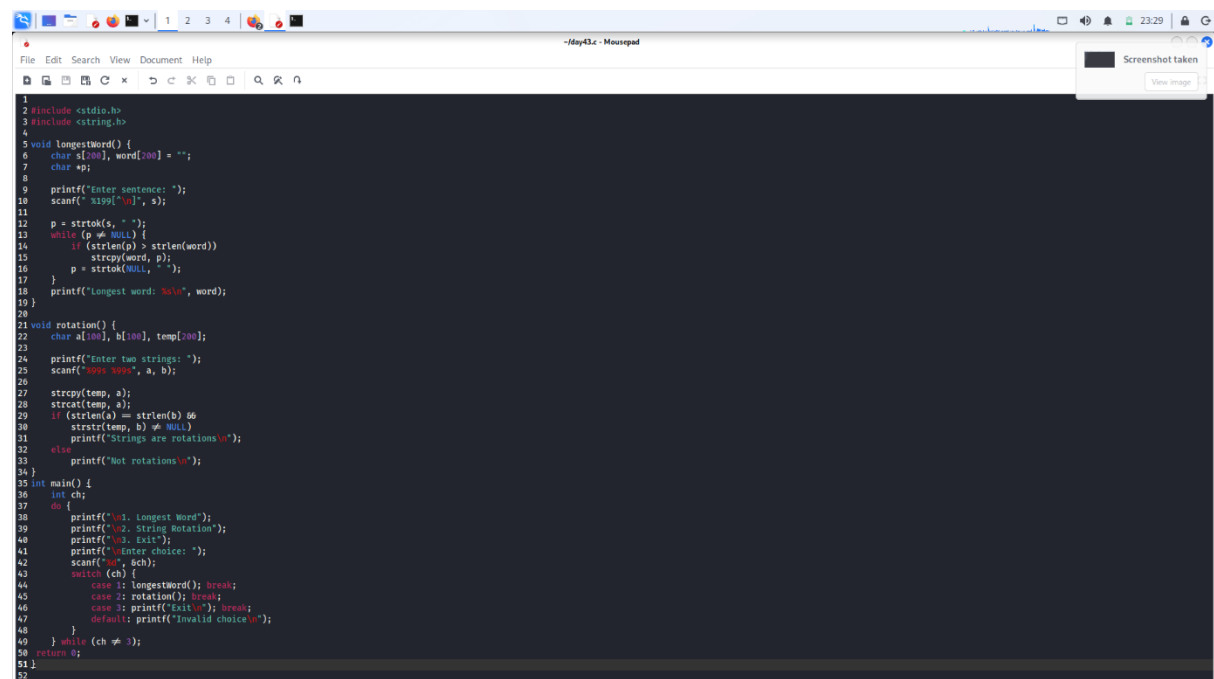
Aim

To write a menu-driven C program to find the longest word in a sentence and check whether two strings are rotations of each other using separate functions.

Algorithm

1. Start the program.
2. Display the menu with two operations and an exit option.
3. Read the user's choice.
4. If choice is 1, find and display the longest word.
5. If choice is 2, check whether the strings are rotations.
6. Repeat until the user chooses Exit.
7. Stop.

>C Program



```
1
2 #include <stdio.h>
3 #include <string.h>
4
5 void longestWord() {
6     char s[200], word[200] = "";
7     char *p;
8
9     printf("Enter sentence: ");
10    scanf("%199[^\n]", s);
11
12    p = strtok(s, " ");
13    while (p != NULL) {
14        if (strlen(p) > strlen(word))
15            strcpy(word, p);
16        p = strtok(NULL, " ");
17    }
18    printf("Longest word: %s\n", word);
19 }
20
21 void rotation() {
22     char a[100], b[100], temp[200];
23
24     printf("Enter two strings: ");
25     scanf("%99[^\n]", a, b);
26
27     strcpy(temp, a);
28     strcpy(temp, b);
29     if (strlen(a) == strlen(b) &&
30         strstr(temp, b) != NULL)
31         printf("Strings are rotations\n");
32     else
33         printf("Not rotations\n");
34 }
35
36 int main() {
37     int ch;
38     do {
39         printf("\n1. Longest Word");
40         printf("\n2. String Rotation");
41         printf("\n3. Exit");
42         printf("\nEnter choice: ");
43         scanf("%d", &ch);
44         switch (ch) {
45             case 1: longestWord(); break;
46             case 2: rotation(); break;
47             case 3: printf("Exit\n"); break;
48             default: printf("Invalid choice\n");
49         }
50     } while (ch != 3);
51     return 0;
52 }
```

>OUTPUT;

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day43.c -o day43
[priyadarshna@kali]~$ ./day43.c
zsh: permission denied: ./day43.c
[priyadarshna@kali]~$ ./day43
1. Longest Word
2. String Rotation
3. Exit
Enter choice: 1
Enter sentence: I love programming
Longest word: programming
1. Longest Word
2. String Rotation
3. Exit
Enter choice: 2
Enter two strings: ABCD CDAB
Strings are rotations
1. Longest Word
2. String Rotation
3. Exit
Enter choice: 3
Exit
[priyadarshna@kali]~$
```